



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (MMD)(CBCS) Degree Examinations
January 2023

22MMD133 - MECHATRONICS SYSTEM DESIGN

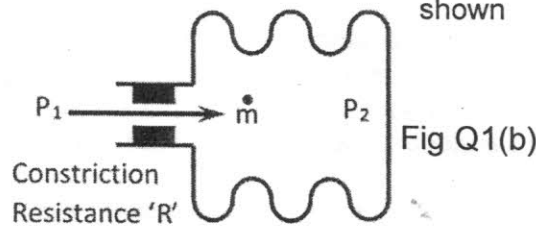
Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

1. a) Obtain the mathematical model for an armature controlled DC motor.
b) Obtain the expression for the pressure variations inside the bellows 'P₂' when there is an input compressed air at a pressure of 'P₁' as shown in Fig.Q1(b)



Marks	BT*	CO*	PO*
10	L*3	1	1

10	L3	2	1
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2. a) A 10kΩ single-turn potentiometer is used to sense the 360degree rotation of a shaft. If the supply voltage to the potentiometer is 10V, what is the output voltage when the shaft rotates through 60 degrees?
b) Derive the mathematical model for the liquid system as shown in the **figure 2(b)** considering the pressure 'p₁' and flow rate 'q₄' as the inputs and output as the flow rate 'q₃'.

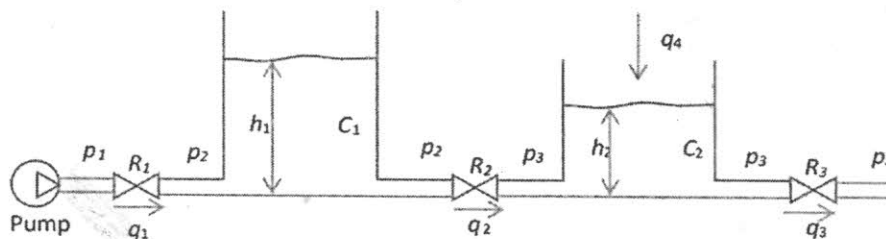


figure 2(b)

10	L3	2	1
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3. a) A platinum resistance temperature sensor has a resistance of 50 Ω at 0° Centigrade and forms one arm of Wheatstone Bridge. The bridge is balanced at this temperature with each of the other arm is being 50Ω. If the temperature coefficient of resistance of platinum is 0.004/K, What will be the output voltage from the bridge per degree change in temperature? Assume the supply voltage as 6V.
b) With the help of a block diagram, explain the steps involved in Mechatronics System Design

10	L3	1	1
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10	L2	2	1
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Unit-II

4. a) Explain the working principle of two phase unipolar permanent magnet stepper motor and write the switching sequence for full stepping and half stepping mode.

10	L3	3	1
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P.T.O.

- b) Design a pneumatic circuit for the sequence of operations mentioned as following: Initially the double acting cylinder is in a completely extended position. When a push button has been pressed, the extended cylinder (1A) retracts completely and executes a to-and-fro movement in the rear stroke range. The oscillating is limited to the retracted end position by a roller lever valve as well as a second roller lever valve in the central position. The frequency of oscillating is adjustable within limits by setting a pressure regulator controlling the amount of air supply. Set an operating pressure of $p = 4 \text{ bar}$ ($= 400 \text{ kPa}$) to accomplish this to-and-fro motion. After a specified interval, the oscillator is switched off. The double-acting cylinder extends completely and actuates the third roller lever valve. Set a vibration time of $t = 5 \text{ seconds}$.

10 L3 4 1

5. a) Parts are to be stamped with a stamping device. By pressing two pushbutton switches the die is pushed down and the part is stamped. When the stamping pressure has been achieved the die is returned to its start position. Design a electro-pneumatic system to implement this operation.

10 L3 3 1

- b) Design an instrumentation amplifier for an amplification ratio of 2500 using $500\text{k}\Omega$, $200\text{k}\Omega$, $100\text{k}\Omega$, $1\text{k}\Omega$ resistors.

10 L3 4 1

6. a) Parts are to be clamped using a clamping device. By pressing a pushbutton switch the moveable clamping jaw is pushed forward and the part is clamped. By pressing another pushbutton switch the clamping jaw is returned to its start position. Design a electro-pneumatic system to implement this operation.

10 L3 3 1

- b) A 4 bit R-2R ladder digital to analog converter has an input reference voltage of 8V. Derive the equation for the output of the R-2R ladder DAC. Obtain the analog output for all the possible combinations of 4 bit digital number.

10 L3 4 1

Unit-III

7. a) With the help of sketches explain the working principle of MEMS pressure sensors. Explain the different methods for sensing the strain in the diaphragm.

10 L2 5 1

- b) Find the unit step response for the system described by the transfer

$$\text{function } G(s) = \frac{4}{s+4} \text{ . Also find the time constant, raise time and}$$

settling time.

10 L3 5 1

8. a) Explain the 4 types of microactuation systems used in MEMS systems

10 L2 5 1

- b) Consider a thermocouple which has a transfer function linking its voltage output V and temperature input of

$$G(s) = \frac{30 \times 10^{-6}}{10s + 1} \text{ V/}^{\circ}\text{C}$$

Determine the response of the system when subject to a step input of size 100°C and the time taken to reach 95% of the steady-state value.

10 L3 5 1